Causal Representation Learning and Causal Discovery in Deep Learning: Foundations, Methodologies, and Frontiers

1 Introduction

The integration of causal inference with deep learning represents a paradigm shift from passive pattern recognition to active mechanistic understanding, addressing fundamental limitations inherent in purely correlational models. While standard deep learning architectures have achieved unprecedented success in static distribution settings, they frequently fail under distribution shifts, rely on spurious correlations, and lack the capacity for counterfactual reasoning [1; 2]. These limitations stem from the fact that observational data alone is insufficient for learning interventional or counterfactual distributions without additional structural assumptions, a constraint formalized by the causal hierarchy theorem which holds even for universally expressive neural networks [3]. Consequently, achieving robustness and generalizability requires moving beyond statistical association to recover the underlying data-generating processes [4].

This survey delineates two primary objectives within this emerging field: causal representation learning and causal discovery. Causal representation learning seeks to disentangle high-dimensional sensory inputs, such as images or text, into latent variables that correspond to true causal factors of variation [5]. Unlike traditional unsupervised learning which often yields statistically independent but causally entangled features, causal representation learning leverages inductive biases such as mechanism independence and multi-environment diversity to ensure identifiability [6; 7]. Recent theoretical advancements demonstrate that without interventional data or specific weak supervision, recovering these latent causal structures remains ill-posed, necessitating frameworks that can exploit geometric signatures left by unknown interventions [8; 9].

Conversely, causal discovery focuses on inferring the graphical structure of causal relationships among variables, transitioning from discrete combinatorial search to continuous optimization paradigms enabled by deep learning [10]. Modern approaches utilize differentiable acyclicity constraints and neural parameterizations to scale causal graph learning to high-dimensional regimes, although challenges regarding scale-invariance and the distinction between parsimonious explanations and true causal mechanisms persist [11; 12]. Furthermore, the synergy between these fields is evident as deep neural networks provide the expressive power to model non-linear causal mechanisms, while causal principles offer the inductive biases necessary for data efficiency and out-of-distribution performance [13].

The organizational structure of this survey reflects this duality and their convergence. We begin by establishing the theoretical foundations and identifiability conditions required to bridge structural causal models with neural architectures. Subsequently, we examine algorithmic paradigms for deep causal structure discovery, including differentiable optimization and amortized inference. We then explore architectures designed specifically for causal representation learning and disentanglement, followed by an analysis of causal generative modeling for counterfactual synthesis and effect estimation. Finally, we synthesize applications across scientific domains and discuss evaluation frameworks essential for trustworthy AI. By traversing from foundational theory to methodologies, this work aims to provide a comprehensive roadmap for developing systems that not only predict but also understand and reason about the world [14].

2 Theoretical Foundations and Identifiability in Latent Causal Spaces

2.1 Formalizing Neural Structural Causal Models and Expressivity

The integration of Structural Causal Models (SCMs) with deep neural architectures establishes a rigorous mathematical framework for modeling non-linear data-generating processes. Formally, an SCM is defined as a tuple $\langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$, where $\mathbf{V}$ represents endogenous variables determined by structural equations $V_i := f_i(\mathrm{PA}_i, U_i)$. In the neural paradigm, the mechanisms f_i and the exogenous noise distributions $P(\mathbf{U})$ are parameterized by deep neural networks, leveraging their universal approximation capabilities to capture complex, high-dimensional dependencies [3]. Architectures such as normalizing flows and autoregressive models have proven particularly effective, as their invertible nature facilitates the bijective mapping between observational data and latent exogenous noise, a prerequisite for counterfactual reasoning [15; 16]. For instance, [17] demonstrate that given a causal ordering, autoregressive flows can identify causal models from observational data, effectively generalizing additive noise models to non-linear settings.

However, a critical distinction must be drawn between expressivity and learnability. While neural networks possess the theoretical capacity to represent any SCM, the Causal Hierarchy Theorem dictates that observational data alone is insufficient to learn interventional or counterfactual distributions without additional structural assumptions [3; 4]. Merely optimizing for likelihood on observational data yields models that are statistically consistent but causally ambiguous, as distinct SCMs can induce identical observational distributions while diverging under intervention. This limitation underscores that standard deep learning objectives, which target the first rung of Pearl's ladder (association), cannot inherently ascend to the second (intervention) or third (counterfactual) rungs [18]. Consequently, the expressivity of neural networks does not guarantee causal identifiability; rather, it necessitates the imposition of specific inductive biases, such as independent causal mechanisms or sparsity constraints, to resolve the inherent ambiguities [19].

Recent theoretical advancements highlight that bridging this gap requires moving beyond purely correlational objectives. [20] propose viewing the network architecture itself as an SCM to compute feature attributions, yet this relies on strong assumptions about the input causal structure. Furthermore, while methods like [21] decouple order search from edge selection in additive settings, extending these guarantees to fully non-linear, neural-parameterized systems remains challenging. The emergence of Neural Causal Models (NCMs) attempts to encode structural constraints directly into the architecture to ensure that the learned functions respect the underlying causal graph [3]. Nevertheless, as noted in [6], invariance principles alone are often insufficient for identifying latent variables without supplementary constraints, suggesting a need for hybrid approaches that combine neural flexibility with rigorous causal regularizers. Future research must therefore focus on developing architectures that inherently embed causal inductive biases, ensuring that the immense expressivity of deep learning is harnessed not just for prediction, but for valid causal inference and robust generalization across distribution shifts.

2.2 Identifiability Theory for Latent Causal Variables

Building upon the neural architectures and structural constraints discussed previously, the identifiability of latent causal variables emerges as the theoretical cornerstone of Causal Representation Learning (CRL). This section addresses the fundamental challenge of recovering high-level causal factors from low-dimensional, non-linear mixtures, moving beyond the expressive capabilities of deep networks to ensure unique recovery. Historically, identifiability was constrained by restrictive parametric assumptions, such as linearity and Gaussianity, which are often violated in complex deep learning domains. Recent theoretical breakthroughs have significantly relaxed these constraints, demonstrating that ground-truth causal structures can be uniquely identified up to trivial indeterminacies—typically permutation and scaling—by leveraging multi-environment diversity, specific intervention types, and non-linear mixing functions.

A primary avenue for achieving non-parametric identifiability involves exploiting distributional shifts across multiple environments. In the absence of interventional labels, observing data generated under varying causal mechanisms allows for the disentanglement of latent factors. [7] establishes that for general non-parametric settings, the observational distribution combined with unknown perfect interventions suffices for identifiability, provided a genericity condition rules out fine-tuned cancellations. This line of inquiry is extended by [22], which shows that under sparsity constraints on the latent graph and sufficient heterogeneity, one can recover the moralized graph of the underlying DAG. Furthermore, [23] provides necessary and sufficient conditions characterizing the types of distribution shifts required for identifiability in latent additive noise models, bridging the gap between theoretical guarantees and practical algorithmic design.

While distributional shifts provide a foundation, the role of explicit interventions remains pivotal in strengthening identifiability guarantees, particularly when environmental diversity is insufficient. While early works focused on known single-node interventions, recent theory addresses the more realistic scenario of unknown and multi-node interventions. [24] proves that given sufficiently diverse interventional environments, identifiability up to ancestors is achievable with soft interventions, while hard interventions permit perfect recovery, effectively matching the guarantees of known single-node settings. Similarly, [25] utilizes score functions to demonstrate that two stochastic hard interventions per node suffice for identifiability even under general non-linear transformations, without requiring knowledge of the specific intervention targets. These results are complemented by [8], which highlights how interventional data carries geometric signatures of the latent support, enabling block affine identification even under imperfect interventions.

Beyond standard mixing assumptions and intervention protocols, emerging theories address the complexities of partial observability and non-invertible generation processes. [26] introduces identifiability results for settings where measurements capture only instance-dependent subsets of the latent state, enforcing sparsity to recover underlying variables. Addressing information loss inherent in sensory data, [27] establishes that temporal context can recover independent latent components even when the instantaneous mixing function is non-invertible. Additionally, [19] proposes mechanism sparsity regularization as a unifying principle, showing that enforcing sparsity in the causal graph relating latents to auxiliary variables or past states facilitates non-parametric identification.

Despite these advances, critical challenges persist regarding the sufficiency of specific inductive biases. [6] warns that invariance alone is insufficient for identifying latent causal variables, necessitating hybrid approaches that combine invariance with structural priors. This limitation underscores the need to explore complementary principles, such as the independence of causal mechanisms, which offers a distinct pathway to disentanglement. Consequently, future research must focus on refining the minimal conditions for identifiability in fully unsupervised, non-stationary regimes and developing scalable algorithms that operationalize these theoretical insights without relying on overly strong faithfulness assumptions, naturally leading to the discussion of the Independent Causal Mechanisms principle in the following section.

2.3 The Principle of Independent Causal Mechanisms and Invariance

The Principle of Independent Causal Mechanisms (ICM) posits that the physical process generating a cause is independent of the mechanism mapping that cause to its effect, serving as a critical inductive bias for disentangling latent representations [28]. This principle implies that the conditional distribution $P(Y|X)$ contains no information about the marginal distribution $P(X)$, a property known as algorithmic independence of conditionals. In the context of representation learning, this independence suggests that causal factors should be modular; changing the mechanism of one factor should not necessitate changes in the mechanisms of others [29]. This modularity has been formalized through Independent Mechanism Analysis (IMA), which extends Independent Component Analysis (ICA) by assuming that sources independently influence the mixing process, thereby resolving non-identifiability issues inherent in nonlinear blind source separation [30]. Furthermore, group-theoretic frameworks have been proposed to unify various interpretations of ICM, assessing the cause-mechanism relationship against null hypotheses via random generic group transformations [31].

A primary operationalization of ICM in deep learning is through invariance, where mechanisms are assumed to remain stable across varying environments or domains. Invariant Risk Minimization (IRM) exemplifies this approach, seeking data representations where the optimal classifier remains invariant across training distributions, theoretically aligning with causal structures [32]. Such invariance is crucial for out-of-distribution generalization, as causal mechanisms are presumed to be transportable across domains unlike spurious correlations [33; 13]. Recent advances have extended these concepts to nonlinear settings, demonstrating that under specific exponential family assumptions, invariant causal representation learning can identify direct causes and guarantee generalization [34]. Similarly, in graph-structured data, capturing invariant subgraphs containing causal information has proven effective for robust learning under distribution shifts [35].

However, the sufficiency of invariance as a sole criterion for causal identification in latent spaces faces significant theoretical challenges. Critical analyses reveal that invariance alone is insufficient to uniquely identify latent causal variables without additional structural constraints, leading to impossibility results in certain settings [6; 36]. Specifically, purely invariant features may fail to recover the true causal graph if the variation across environments does not sufficiently excite all causal directions or if the invariance principle captures non-causal stable dependencies. Consequently, recent research advocates for hybrid approaches that combine invariance with other priors, such as mechanism sparsity. The principle of disentanglement via mechanism sparsity regularization leverages the assumption that latent factors depend sparsely on past factors or auxiliary variables, providing identifiability guarantees even when invariance is weak or absent [19; 37]. Additionally, the integration of interventional data offers geometric signatures that can break symmetries unresolvable by observational invariance alone, enabling provable identification of latent factors under general nonlinear mixing [8; 38]. Future directions must therefore focus on synthesizing invariance with sparsity, temporal dynamics, and interventional geometry to construct robust frameworks for causal representation learning that transcend the limitations of purely observational or single-principle approaches.

2.4 Identifiability in Temporal and Dynamic Causal Systems

Temporal data offers a unique vantage point for causal representation learning, directly addressing the limitations of purely observational or single-principle approaches highlighted previously by leveraging the arrow of time as a powerful inductive bias. In dynamical systems, the assumption that causes must precede effects allows for the disentanglement of latent factors even when the instantaneous mixing function is non-invertible or information-lossy. Recent theoretical advances demonstrate that by leveraging time-delayed dependencies, one can recover independent latent components from nonlinear mixtures, effectively bypassing the strict invertibility requirements of traditional Independent Component Analysis (ICA) [27]. This temporal context acts as a regularizer, enabling the identification of latent causal processes where static methods fail due to the entanglement of shared sources.

A critical distinction in temporal causal discovery lies in separating lagged influences from contemporaneous interactions. While lagged relations are often identifiable through Granger causality principles, contemporaneous effects introduce cyclic dependencies that complicate structural recovery. Extending linear non-Gaussian models to cyclic settings, researchers have utilized Independent Subspace Analysis to exactly identify local directed structures and causal strengths within Markov blankets, even in the presence of feedback loops [39]. Furthermore, frameworks like iCITRIS address the challenge of instantaneous effects in intervened temporal sequences, proving that observing intervention targets—such as agent actions—allows for the simultaneous identification of multidimensional causal variables and their instantaneous graph structure [40]. When intervention targets are unknown, methods leveraging heterogeneous interventional time-series data can mutually promote the recovery of both the Granger causal structure and the specific nodes subjected to imperfect interventions [41].

The role of non-stationarity and distribution shifts further strengthens these identifiability guarantees, echoing the invariance principles discussed earlier but applied to dynamic transitions. Theories positing that latent causal processes exhibit fixed transition dynamics while undergoing distributional shifts enable the factorization of unknown changes into transition distribution variations and observation noise [42; 43]. By enforcing sparsity constraints on the causal graph governing these temporal dependencies, it is possible to achieve nonparametric identifiability of latent factors up to permutation, a principle formalized as disentanglement via mechanism sparsity regularization [19; 37]. Specifically, the TDRL framework illustrates how time-delayed latent causal influences can be reliably identified under stationary environments and distinct distribution shifts, outperforming baselines that ignore the modular representation of temporal changes [44].

However, challenges remain regarding the assumption of exchangeability and the handling of complex, evolving mechanisms. While traditional approaches rely on i.i.d. assumptions, emerging "causal de Finetti" theorems provide a statistical formalization for identifying invariant causal structures in exchangeable data, relaxing the need for strict independence [45]. Additionally, amortized causal discovery frameworks suggest that sharing dynamics across different samples from varying underlying graphs can significantly improve sample efficiency and robustness to hidden confounding in time-series data [46]. These advances in establishing temporal identifiability and recovering dynamic latent structures provide the necessary foundation for the next critical step: mapping these low-level micro-variables to high-level macro-variables through causal abstraction, thereby enabling symbolic reasoning and interpretable counterfactuals.

2.5 Causal Abstraction and Hierarchical Identifiability

Causal abstraction provides the theoretical machinery to map low-level micro-variables, such as pixel intensities or neural activations, to high-level macro-variables that support symbolic reasoning while preserving causal semantics. This hierarchical bridging is essential for transforming opaque deep learning systems into interpretable models capable of counterfactual reasoning at varying granularities. Formally, a high-level causal model $\mathcal{M}_H$ is a valid abstraction of a low-level model $\mathcal{M}_L$ if an alignment function τ exists such that interventional distributions in $\mathcal{M}_H$ commute with those in $\mathcal{M}_L$ under specific transformation operators. Recent advances have shifted from brute-force search methods to learnable neural frameworks. For instance, [47] proposes clustering variables and their domains to construct abstraction functions compatible with Neural Causal Models, enabling tractable inference across the causal hierarchy in high-dimensional image settings. Similarly, [48] utilizes interchange interventions to experimentally verify whether internal neural representations align with predefined symbolic causal structures, demonstrating that sophisticated models like BERT can encode compositional causal logic absent in simpler baselines.

A critical challenge in this domain is ensuring consistency across Pearl's three levels of causation—association, intervention, and counterfactuals. [49] extends abstraction theory to cyclic structures and typed variables, defining approximate causal abstraction to quantify the fidelity of high-level explanations against low-level dynamics. This work notably decomposes constructive abstraction into marginalization, variable-merge, and value-merge operations, providing a rigorous algebraic foundation for simplifying complex systems without losing causal validity. Furthermore, [50] addresses the limitation of disjoint neuron alignments by introducing Distributed Alignment Search (DAS). By leveraging gradient descent over non-standard bases, DAS identifies distributed neural representations that faithfully realize high-level causal variables, overcoming the rigidity of earlier one-to-one mapping assumptions.

Hierarchical identifiability further complicates abstraction, particularly when latent variables form multi-layered structures with unobserved confounders. [51] tackles scenarios where only leaf nodes are measured, employing rank deficiency constraints to locate latent variables and determine their cardinalities within a hierarchical graph. This approach proves that under specific structural restrictions, the Markov equivalence class of the entire latent hierarchy is asymptotically identifiable. Complementing this, [52] introduces Causal Bisimulation Modeling (CBM) to derive task-specific abstractions that retain only causally relevant state variables. By learning implicit causal dynamics, CBM achieves near-oracle sample efficiency, illustrating how hierarchical abstraction facilitates generalization in sequential decision-making.

Despite these advances, significant trade-offs remain between abstraction granularity and identifiability. While [53] guarantees that a target causal model becomes a faithful abstraction of a neural network when Interchange Intervention Training (IIT) loss is zero, achieving this in fully unsupervised settings remains elusive without strong inductive biases. Moreover, the tension between compressing information and preserving counterfactual validity necessitates careful regularization. Future research must focus on developing unified frameworks that jointly learn the abstraction map and the underlying causal graph, particularly in non-parametric settings where functional forms are unknown. Integrating these abstraction principles with foundation models could enable zero-shot causal adaptation, allowing agents to reason about novel environments by manipulating high-level conceptual variables rather than raw sensory inputs.

3 Algorithmic Paradigms for Deep Causal Structure Discovery

3.1 Continuous Optimization and Differentiable Structure Learning

The paradigm of causal structure learning has undergone a transformative shift from discrete combinatorial search to continuous optimization, enabling the application of gradient-based methods to infer Directed Acyclic Graphs (DAGs). Traditionally, score-based methods faced an NP-hard search space superexponential in the number of variables, relying on heuristic local searches that often stagnated in suboptimal solutions [54]. A seminal breakthrough reformulated this problem as a continuous constrained optimization task, introducing a differentiable acyclicity constraint based on the trace of the matrix exponential, thereby allowing the use of standard numerical optimizers [54; 54]. This relaxation permits the parameterization of causal mechanisms via deep neural networks, capturing complex non-linear relationships that linear Structural Equation Models (SEMs) cannot represent [54].

Central to this approach is the integration of expressive neural architectures with structural constraints. For instance, [54] employs a variational autoencoder parameterized by a Graph Neural Network to learn DAGs for both continuous and discrete variables, demonstrating superior accuracy in non-linear settings compared to traditional least-squares approaches. Similarly, [55] leverages the Gumbel-Softmax trick to approximate binary adjacency matrices, facilitating end-to-end training while maintaining near-discrete edge predictions. To address the computational burden of explicit acyclicity penalties, recent works have explored implicit guarantees. [12] proposes ENCO, which decouples edge existence from orientation, ensuring acyclicity by design through independent edge likelihood optimization, thus scaling to hundreds of variables without augmented Lagrangian schemes. Furthermore, autoregressive normalizing flows have been identified as intrinsically causal; their architectural ordering naturally enforces a topological sort, allowing for identifiable causal discovery without explicit constraints [15; 17].

Despite these advances, challenges regarding scalability and robustness persist. While methods like [56] demonstrate near-linear training time scaling, the reliance on differentiable proxies for acyclicity can sometimes lead to numerical instability or convergence to graphs with minor cycles in high-dimensional regimes. To mitigate this, reinforcement learning has been adopted as a flexible search strategy within the continuous framework, where agents learn to construct DAGs by optimizing rewards that balance scoring functions with acyclicity penalties [57; 58]. Additionally, the incorporation of interventional data has proven critical for resolving identifiability issues inherent in observational settings. [59] extends the continuous framework to leverage interventional distributions, utilizing normalizing flows to handle unknown intervention targets effectively.

Emerging trends indicate a move toward Bayesian formulations and amortized inference to quantify epistemic uncertainty, which point estimates often obscure. Methods such as [60] and [61] utilize variational inference over relaxed graph spaces to approximate posterior distributions over DAGs. Moreover, the integration of score matching techniques offers a promising direction for scalable discovery in additive noise models by utilizing the Hessian of the log-density to recover topological orders efficiently [62; 63]. Future research must address the tension between the flexibility of non-parametric neural mechanisms and the stringent identifiability conditions required for valid causal interpretation, particularly in the presence of latent confounders where differentiable algebraic constraints on Acyclic Directed Mixed Graphs (ADMGs) are beginning to emerge [64].

3.2 Neural Amortization and Bayesian Causal Discovery

Building upon the Bayesian formulations introduced to quantify epistemic uncertainty, neural amortization represents a paradigmatic shift in causal discovery, transitioning from computationally intensive, instance-specific search procedures to learning a direct mapping from data distributions to causal structures. Traditional score-based or constraint-based methods typically require solving a combinatorial optimization problem for every new dataset, a process that scales poorly with dimensionality and sample scarcity. In contrast, amortized approaches train a neural inference network f_ϕ on a meta-distribution of synthetic causal graphs, enabling the instantaneous prediction of posterior distributions over Directed Acyclic Graphs (DAGs) for unseen data. This strategy not only accelerates inference by orders of magnitude but also allows the model to internalize domain-specific inductive biases, facilitating robust generalization to distribution shifts often encountered in real-world applications [65].

A primary implementation of this framework utilizes Variational Autoencoders (VAEs) to approximate the intractable posterior $P(G|D)$ over graph structures G given data D. By encoding observational or interventional datasets into a latent representation, these models decode parameters defining a variational distribution over adjacency matrices. Notably, [66] pioneers this by parameterizing the variational family with Graph Neural Networks, effectively capturing complex non-linear dependencies that linear Structural Equation Models miss. Similarly, [60] advances scalable variational inference by employing continuous relaxations of the discrete graph space, ensuring low-variance stochastic optimization while maintaining acyclicity through expressive variational families. These methods excel in low-data regimes where point estimates fail to capture structural uncertainty, a critical limitation of maximum-likelihood approaches.

Beyond variational inference, Generative Flow Networks (GFlowNets) have emerged as a powerful tool for Bayesian causal discovery. Unlike VAEs that optimize a lower bound, GFlowNets are trained to sample DAGs with probability proportional to their posterior reward, thereby providing a more accurate characterization of the multi-modal posterior landscape inherent in causal discovery. The integration of GFlowNets with Variational Bayes, as seen in [67], enables the joint learning of graph structure and causal mechanisms, offering guarantees of acyclic sampling without the need for complex augmented Lagrangian constraints required by differentiable DAG learning baselines [68]. This capability is particularly vital when dealing with non-identifiable structures where multiple graphs explain the data equally well.

The efficacy of amortized methods extends to temporal domains, where shared dynamical principles across different time-series instances can be leveraged. [46] demonstrates that training a single model to infer causal relations across samples with varying underlying graphs significantly outperforms fitting individual models, particularly under noise and hidden confounding. Furthermore, recent innovations propose "foundation model" approaches for causal discovery, where deep learning models are pretrained to resolve predictions from classical algorithms on variable subsets, achieving inference speeds orders of magnitude faster than existing models while maintaining state-of-the-art accuracy [14].

Despite these advances in scalability and speed, challenges remain regarding the "reality gap" between synthetic training data and real-world distributions, particularly when the assumption of causal sufficiency is violated. While amortized models show remarkable robustness to distribution shifts [65], their performance hinges on the diversity of the synthetic meta-training distribution and often struggles when unmeasured common causes are present. Consequently, future research must address the theoretical guarantees of identifiability within amortized frameworks and explore hybrid architectures that combine the speed of neural amortization with the rigorous constraints of symbolic causal calculus. Such developments are essential to ensure both efficiency and correctness in high-stakes domains, paving the way for robust causal discovery in the presence of latent confounders, a topic explored in the following section.

3.3 Causal Discovery in the Presence of Latent Confounders

The presence of unmeasured common causes, or latent confounders, represents a fundamental barrier to valid causal discovery, often inducing spurious correlations that mislead standard algorithms assuming causal sufficiency. In the era of deep learning, addressing this challenge requires moving beyond discrete constraint-based searches toward continuous, differentiable frameworks capable of jointly inferring latent structures and observed causal mechanisms. A primary paradigm involves leveraging deep generative models to approximate the posterior distribution of unobserved confounders. For instance, variational autoencoders (VAEs) have been adapted to treat latent variables as stochastic nodes within a Structural Causal Model (SCM), optimizing evidence lower bounds to simultaneously recover confounder distributions and causal graphs [69]. This approach allows for the integration of proxy variables, where high-dimensional unstructured data, such as images or text, serve as noisy measurements of hidden confounders, significantly improving adjustment accuracy in complex domains like genomics and healthcare [70].

Complementing generative approaches, recent advances have extended independence criteria to non-linear settings through residual analysis. The Generalized Independent Noise (GIN) condition provides a rigorous theoretical foundation for identifying latent causal graphs by testing the independence between specific linear combinations of observed variables and their residuals, effectively generalizing traditional independent noise conditions to scenarios with latent confounding [71]. Building on this, differentiable causal discovery methods have emerged to handle the combinatorial complexity of Acyclic Directed Mixed Graphs (ADMGs), which encode both direct causal effects and bidirected edges representing latent confounding. By deriving differentiable algebraic constraints that characterize the space of ancestral and bow-free ADMGs, researchers can now employ gradient-based optimization to learn non-linear functional relations in confounded systems, overcoming the scalability limitations of discrete search procedures [64; 72]. These methods are particularly potent when combined with autoregressive flows, enabling the simultaneous estimation of structural relationships and treatment effects [72].

Furthermore, the heterogeneity of confounding effects necessitates models that can distinguish between global and local confounding structures. Mixture models and clustering techniques have been proposed to address class-specific confounding, where latent variables induce distinct causal regimes across subpopulations [73]. In networked data, hidden confounders often manifest through structural dependencies; frameworks like the network deconfounder exploit graph topology to infer latent patterns of confounding, thereby refining individual causal effect estimates [74]. Similarly, the "deconfounder" algorithm leverages multi-cause settings to construct substitute confounders via unsupervised learning, offering a checkable approach to causal inference that relaxes the strict ignorability assumption [75]. Despite these advances, critical challenges remain regarding identifiability. Recent theoretical work highlights that without specific parametric assumptions or interventional data, deep latent variable models may yield inconsistent causal estimates due to the inherent non-identifiability of the underlying SCM [76]. Future research must therefore focus on hybrid strategies that combine the expressive power of deep generative models with rigorous identifiability conditions derived from multi-environment data or known structural constraints to ensure robust causal discovery in fully observed and latent settings alike [23].

3.4 Temporal Dynamics and Instantaneous Causal Inference

Building on the necessity for hybrid strategies that combine deep generative models with rigorous identifiability conditions in latent settings, as highlighted in the previous section, temporal dynamics offer a unique structural asymmetry that often resolves such ambiguities. However, distinguishing between lagged dependencies and instantaneous effects remains a formidable challenge when sampling rates are coarser than the underlying causal timescales. Traditional approaches relying on Vector Autoregression (VAR) or standard Granger causality frequently fail to capture non-linear interactions or contemporaneous confounding, necessitating the integration of deep neural architectures capable of modeling complex dynamical systems. Recent advancements have shifted towards frameworks that jointly learn latent causal processes and their structural relations, leveraging temporal order to disentangle immediate from delayed influences. For instance, the Rhino framework combines variational inference with deep learning to model non-linear relationships characterized by history-dependent noise and instantaneous effects, theoretically proving structural identifiability under these relaxed conditions [77]. Similarly, LEAP extends Variational Autoencoders to recover time-delayed latent causal variables from general temporal data by enforcing non-stationarity and transition constraints, offering a robust solution for unsupervised identification of latent dynamics [42].

A critical frontier in this domain is the treatment of instantaneous causality, which arises when the measurement interval exceeds the speed of causal propagation. While early identifiability results assumed purely time-delayed interactions, newer methods like iCITRIS explicitly address instantaneous effects in intervened temporal sequences, utilizing differentiable causal discovery to learn multidimensional causal variables and their graphs simultaneously [40]. Complementing this, the IDOL framework imposes sparse influence constraints on both time-delayed and instantaneous relations, establishing identifiability through sufficient variability and contextual information without requiring explicit intervention targets [78]. These approaches contrast with constraint-based methods like PCMCI+, which optimize conditioning sets to improve the detection of contemporaneous links in autocorrelated time series, demonstrating higher recall and better false positive control compared to traditional PC algorithms [79].

The integration of attention mechanisms and Graph Neural Networks (GNNs) has further revolutionized temporal causal inference by enabling the modeling of long-range dependencies and complex spatial-temporal interactions. Transformer-based architectures, such as the Causal Transformer, utilize cross-attention mechanisms to capture intricate dependencies among time-varying confounders for counterfactual estimation, outperforming recurrent baselines in longitudinal settings [80]. In the context of Granger causality, self-explaining neural networks and Jacobian regularizer-based approaches have been developed to infer interaction signs and full-time causal matrices within a single unified model, addressing the scalability limitations of variable-specific predictive models [81; 82]. Furthermore, amortized causal discovery frameworks leverage shared dynamics across different samples to infer causal graphs from time-series data more efficiently, learning a single model that generalizes across varying underlying structures [46].

Despite these advances, significant challenges persist regarding non-stationarity and latent confounding in dynamic systems, creating a complex landscape where statistical inference alone often falls short. Methods like LiLY address distribution shifts by factorizing unknown changes into transition dynamics and global observation shifts, allowing for rapid model correction with few samples [43]. However, the interplay between instantaneous effects and unmeasured confounders often requires strong assumptions about sparsity or noise distributions. To overcome these limitations, future research must look beyond purely data-driven approaches and explore the integration of external knowledge sources. This necessitates relaxing current assumptions by combining interventional data with temporal observations to achieve provable identification in fully non-parametric settings. Such efforts will bridge the gap between theoretical identifiability and practical applicability, naturally leading to the emerging paradigm of leveraging Large Language Models to inject semantic priors and resolve ambiguities in causal structure learning, as discussed in the following section.

3.5 Large Language Models as Priors and Orchestrators in Causal Discovery

The integration of Large Language Models (LLMs) into causal discovery represents a paradigm shift from purely data-driven statistical inference to knowledge-augmented structural learning. While traditional algorithms rely exclusively on observational distributions to infer Directed Acyclic Graphs (DAGs), they often struggle with identifiability in finite-sample regimes or when latent confounders obscure true mechanisms. LLMs address these limitations by serving as orchestrators that inject semantic priors derived from vast corpora of scientific literature and common sense, effectively constraining the combinatorial search space of causal graphs. Recent frameworks leverage LLMs not merely as query tools but as active components in the discovery pipeline, utilizing their ability to reason over variable metadata to propose initial causal skeletons or validate edges inferred by statistical methods [83]. This synergy is formalized in approaches where LLMs provide "knowledge-based" priors that are subsequently refined by data-driven score functions, creating a hybrid baseline that outperforms either modality in isolation [84].

A critical application of LLMs lies in their capacity to act as error-resilient refiners within autonomous discovery loops. Although LLMs possess extensive world knowledge, they are susceptible to hallucinations and logical fallacies, such as inferring causation solely from temporal precedence or textual co-occurrence [85; 86]. To mitigate these risks, emerging methodologies like the Autonomous LLM-Augmented Causal Discovery Framework (ALCM) implement iterative feedback mechanisms where data-driven algorithms critique and correct LLM-generated hypotheses, ensuring the final graph adheres to both statistical constraints and acyclicity requirements [87]. Furthermore, specific strategies have been developed to detect and rectify "order-reversed" errors introduced by LLM priors by analyzing their impact on quasi-circular structures within the graph, thereby enhancing robustness without requiring human intervention [88]. In temporal domains, LLMs facilitate the extraction of meta-knowledge from unstructured system logs to guide meta-initialization, enabling causal discovery in industrial settings where interventional targets are unavailable [89].

Beyond direct graph construction, LLMs are increasingly employed to uncover high-level hidden variables from unstructured data, a task traditionally reliant on expert-defined features. Frameworks such as COAT (Causal representatiOn AssistanT) utilize LLMs as factor proposers that parse raw text into structured causal factors, which are then fed into rigorous constraint-based learners like FCI to recover the underlying causal system [90]. This capability extends to small language models (SLMs), where prompting strategies enriched with knowledge graph structures allow resource-constrained models to surpass larger counterparts in knowledge-based causal discovery tasks [91]. Despite these advances, challenges remain regarding the reliability of LLMs in novel domains where pre-training correlations may not reflect true causal mechanisms, necessitating careful calibration between linguistic priors and empirical evidence [92]. Future research must focus on developing neuro-symbolic architectures that formally integrate the probabilistic reasoning of causal models with the semantic density of LLMs, moving toward causal foundation models capable of zero-shot adaptation across diverse scientific disciplines [93].

4 Architectures for Causal Representation Learning and Disentanglement

4.1 Causal Generative Architectures with Structural Priors

The integration of Structural Causal Model (SCM) constraints into deep generative architectures represents a pivotal shift from learning statistically independent latent factors to recovering causally disentangled representations. Traditional variational autoencoders (VAEs) often rely on the assumption of prior independence among latent variables, which fails to capture the inherent dependencies dictated by underlying causal mechanisms. To address this, recent frameworks explicitly encode causal graphs as structural priors within the generative process. For instance, [94] proposes utilizing an SCM as the prior distribution for bidirectional generative models, enabling the disentanglement of causally related factors even when traditional independence assumptions are violated. Similarly, [95] introduces a Structural caUsAl Variational autoEncoder that learns latent causal representations and their interrelationships simultaneously, proving asymptotic consistency in recovering true parameters under linear-Gaussian assumptions with modulated noise.

Normalizing flows offer a complementary pathway by leveraging their bijective nature to facilitate tractable counterfactual inference, a capability often absent in standard VAEs. The intrinsic correspondence between autoregressive flow architectures and identifiable causal models allows for the recovery of causal directions based on fixed variable orderings. [15] demonstrates that affine and additive autoregressive flows can identify causal models and evaluate interventional queries directly, generalizing additive noise models to high-dimensional settings. This approach is further refined in [17], which establishes that causal models are identifiable from observational data given a causal ordering, providing a rigorous mechanism to implement the do-operator within flow-based architectures. Moreover, [16] combines normalizing flows with variational inference to enable the abduction of exogenous noise variables, a critical step for generating valid counterfactuals in imaging applications that satisfies all three levels of Pearl's causal hierarchy.

Despite these advances, significant challenges remain regarding identifiability in the presence of unobserved confounders and non-invertible mixing functions. While [69] utilizes VAEs to estimate latent confounders from proxy variables, ensuring the uniqueness of the recovered causal structure often requires interventional data or specific environmental diversity. [8] highlights that interventional data carries geometric signatures that can break dependencies between latent factors and their ancestors, enabling provable identification up to permutation and scaling without restrictive distributional assumptions. Furthermore, the issue of non-invertibility in temporal settings is addressed by [27], which establishes identifiability theories for recovering independent latent components from nonlinear, non-invertible mixes by exploiting temporal context.

Looking forward, the synthesis of mechanism sparsity regularization with generative modeling offers a promising direction for partial disentanglement in nonparametric settings. [19] and its extension [37] formalize how sparsity in causal mechanisms can serve as a powerful inductive bias for identifying latent factors without requiring full knowledge of the intervention targets. Future architectures must increasingly bridge the gap between flexible deep generative modeling and rigorous causal identifiability conditions, moving towards frameworks that can robustly handle unknown intervention targets and complex, cyclic dependencies while maintaining the capacity for high-fidelity counterfactual synthesis.

4.2 Invariant Representation Learning for Out-of-Distribution Generalization

Building upon the theoretical groundwork for identifying latent factors via mechanism sparsity and generative modeling, invariant representation learning emerges as a pivotal architectural paradigm for achieving out-of-distribution (OOD) generalization by isolating stable causal mechanisms from spurious, environment-specific correlations. The foundational premise posits that while statistical associations may fluctuate across domains due to distribution shifts, the underlying causal mechanisms generating the target variable remain invariant. Invariant Risk Minimization (IRM) formalizes this objective by seeking a data representation such that the optimal predictor atop it remains constant across all training environments [32]. However, theoretical scrutiny reveals significant limitations; [36] demonstrates that in non-linear regimes, IRM can fail catastrophically unless test distributions are sufficiently similar to training data, often performing no better than standard Empirical Risk Minimization. This necessitates more sophisticated architectural strategies that go beyond simple gradient matching, bridging the gap between abstract invariance principles and the concrete generative mechanisms discussed previously.

To address these failures, recent architectures explicitly decompose latent spaces into causal and non-causal factors, moving closer to the disentangled representations enabled by SCM-constrained generative models. The Causal Semantic Generative model (CSG) employs variational inference to separate semantic factors (causes of the label) from variation factors, proving that identifying the semantic factor bounds the OOD generalization error even from a single training domain [96]. Similarly, [13] proposes enforcing properties of separation, joint independence, and causal sufficiency to simulate invariant causal mechanisms, thereby mitigating the reliance on superficial statistical dependencies. In the domain of visual recognition, [33] argues that standard classifiers fail because image-label associations are not transportable, whereas the causal effect is; their approach estimates this invariant causal effect using deep representations as proxies for unobserved confounders.

The challenge extends to complex structured data, where graph neural networks (GNNs) often exploit spurious structural correlations, requiring adaptations of the causal priors established in earlier sections. [97] introduces a causal variable distinguishing regularizer to correct biased training distributions, forcing the model to focus on stable subgraph patterns rather than environment-sensitive features. Complementing this, [98] conducts interventions on the training distribution to filter out unstable patterns, constructing intrinsically interpretable models that maintain performance under shift. Furthermore, [99] advances this by simultaneously incorporating label and environment causal independence through adversarial training, providing theoretical guarantees for causal subgraph discovery.

Emerging trends also leverage meta-learning and transformation invariance to relax strict domain partition requirements, echoing the sparsity principles noted in the transition toward self-supervised learning. [100] utilizes the speed of adaptation to distributional changes as a signal to learn modular causal structures, assuming that genuine causal mechanisms undergo sparse shifts. Meanwhile, [101] proposes a minimax optimal approach that learns from transformations modifying non-causal features while leaving causal parts unchanged, theoretically enabling OOD generalization even with limited domain diversity. Despite these advances, [6] provides a critical impossibility result, showing that invariance alone is insufficient to identify latent causal variables without additional structural constraints. Consequently, future architectures must hybridize invariance principles with explicit causal graph learning or interventional data to ensure robust identification of the true generative factors, naturally leading to the self-supervised approaches that leverage temporal dynamics and non-stationarity to satisfy these missing constraints.

4.3 Self-Supervised and Temporal Causal Disentanglement

Self-supervised causal representation learning addresses the fundamental challenge of recovering latent causal factors from high-dimensional observations without explicit intervention labels or ground-truth graphs. This paradigm leverages inherent structural priors in data, specifically temporal dynamics, non-stationarity, and sparse mechanism changes, to satisfy identifiability conditions that are otherwise unattainable in static, i.i.d. settings. A primary avenue involves exploiting temporal dependencies to disentangle causally related latent processes. Frameworks such as TDRL and LiLY utilize time-delayed causal influences and distribution shifts in sequential data to identify nonparametric latent causal processes [44; 43]. These methods operate on the principle that while the mixing function from latent to observed space may be complex, the transition dynamics in the latent space often exhibit modularity and sparsity. By enforcing constraints on the transition graph, these models can recover latent variables up to trivial indeterminacies, even when the observation process is non-invertible, as demonstrated by CaRiNG, which utilizes temporal context to mitigate information loss inherent in dimensionality reduction [27]. Furthermore, recent advances have extended these identifiability guarantees to settings with instantaneous effects, overcoming the limitation of purely lagged assumptions through sparse influence constraints and contextual variability [78].

Beyond temporal structure, self-supervised approaches increasingly rely on the principle of mechanism sparsity and multi-environment diversity. Theoretically, if latent factors depend sparsely on past factors or auxiliary variables, regularizing the learned causal graph for sparsity can induce disentanglement without requiring known intervention targets [19; 37]. This connects closely with meta-learning objectives, where the speed of adaptation to distributional changes serves as a proxy for identifying modular causal mechanisms [100]. In scenarios where data arises from multiple domains or non-stationary environments, methods like CITRIS and iCITRIS exploit the geometric signatures of interventions in temporal sequences to identify scalar and multidimensional causal factors, even when only partial components are affected [102; 40]. Additionally, contrastive learning strategies have been adapted to uncover latent causal variables by analyzing the high-dimensional geometry of data distributions resulting from unknown single-node interventions, effectively bypassing the need for paired counterfactual data [38].

Despite these advances, significant challenges remain. While temporal and non-stationary signals provide strong inductive biases, theoretical impossibility results suggest that invariance alone is insufficient for identifying latent causal variables without additional structural constraints [6]. Moreover, most current methods assume a degree of observability regarding intervention targets or specific forms of non-stationarity that may not hold in fully unsupervised real-world applications. Future research must therefore focus on relaxing these assumptions, perhaps by integrating causal abstraction hierarchies or leveraging multi-view partial observability to further constrain the solution space [103; 26]. The synthesis of temporal dynamics with mechanism sparsity offers a promising path toward truly unsupervised causal discovery, enabling agents to reason about causes and effects in complex, evolving environments.

4.4 Hierarchical Causal Abstractions and Object-Centric Representations

Building on the foundational progress in recovering flat latent factors through temporal and non-stationary signals, the field must now address a critical limitation: standard disentangled models often yield statistically independent variables that fail to capture the compositional or hierarchical structure inherent in complex data-generating processes. The transition from these flat representations to hierarchical causal abstractions represents a pivotal evolution, enabling systems to reason across multiple levels of granularity. Central to this endeavor is the formalization of causal abstraction, which defines the conditions under which a high-level structural causal model (SCM) faithfully simplifies a low-level neural system while preserving interventional consistency. Geiger et al. [48] proposed a rigorous framework utilizing interchange interventions to verify whether neural representations align with variables in an interpretable causal model, demonstrating that high-performing transformers can internalize compositional logical structures. Building on this, Beckers et al. [49] generalized the theory to cyclic structures and introduced approximate causal abstraction, allowing for the quantitative assessment of how well high-level explanations map to opaque low-level dynamics.

A primary architectural challenge lies in learning these abstraction functions automatically rather than prescribing them a priori. Xia et al. [47] addressed this by developing a learnable framework that clusters variables and domains to construct abstractions consistent with Pearl's causal hierarchy, effectively bridging the gap between microscopic neural activations and macroscopic causal concepts. Similarly, Rischel et al. [104] introduced a differentiable programming approach to jointly learn abstractions that maintain consistency across multiple interventional distributions, a necessity for robust reasoning in dynamic environments. These methods collectively suggest that valid abstractions must satisfy a commutativity condition between the abstraction map and the causal mechanisms, ensuring that interventions at the macro-level yield predictions consistent with those derived from the micro-level.

Parallel to the development of hierarchical abstractions, object-centric representation learning seeks to decompose scenes into discrete, causally independent entities, mirroring human perceptual organization and providing the semantic units necessary for such hierarchies. Traditional slot-based architectures often struggle with the non-injectivity of multi-object generative functions and the binding problem. Kiciman et al. [105] highlighted that models leveraging object-centric priors significantly outperform distributed counterparts in interventional downstream tasks, yet identifying these latent structures remains challenging without strong supervision. To address the efficiency and scalability of object discovery, Greff et al. [106] proposed the Complex AutoEncoder, which utilizes phase differences in complex-valued activations to bind features into objects without explicit slot mechanisms, offering a biologically plausible and computationally efficient alternative. Furthermore, the integration of causal induction with attention mechanisms has proven vital for goal-directed tasks. Wang et al. [107] demonstrated that agents can incrementally generate causal graphs from visual observations using attention, selectively utilizing induced structures to generalize to novel environments with unseen causal topologies.

Despite these advances, significant trade-offs persist between the expressivity of hierarchical models and their identifiability. While methods like [51] provide theoretical guarantees for identifying latent hierarchies via rank constraints, extending these to non-linear, deep neural settings with partial observability remains an open frontier. Future research must focus on unifying object-centric decomposition with hierarchical causal abstraction, potentially through neuro-symbolic architectures that enforce modular decomposability while learning continuous abstraction maps. Such integration is essential for developing causal foundation models capable of zero-shot adaptation and counterfactual reasoning in complex, multi-scale physical worlds, thereby laying the necessary groundwork for the symbolic causal reasoning and interpretable hybrid paradigms explored in the subsequent section.

4.5 Neuro-Symbolic and Interpretable Causal Embeddings

The integration of symbolic causal reasoning with neural representation learning constitutes a pivotal advancement in achieving interpretable and verifiable artificial intelligence. This hybrid paradigm addresses the opacity of deep learning by aligning distributed neural activations with explicit, human-understandable causal variables, thereby enabling rigorous counterfactual analysis and mechanistic explanation. A primary architectural strategy involves the construction of Causal Concept Embedding Models (Causal CEMs), where decision-making processes are mediated by explicit concept vectors rather than latent black-box features. For instance, [108] demonstrates how Variational Autoencoders can be employed to estimate the causal effect of human-interpretable concepts on predictions, effectively disentangling genuine causal drivers from spurious correlations that plague purely associational models. Similarly, [109] extends this logic to natural language processing by fine-tuning contextual embeddings with adversarial tasks derived from causal graphs, allowing for the estimation of true concept effects in high-dimensional text data.

Beyond concept-based explanations, recent methodologies leverage the theory of causal abstraction to formally verify the alignment between low-level neural computations and high-level symbolic models. The framework of Interchange Intervention Training (IIT), as proposed in [53], trains neural networks to match the counterfactual behavior of a target symbolic causal model, guaranteeing that the neural architecture realizes the intended causal abstraction when convergence is achieved. This approach overcomes the limitations of brute-force search by utilizing gradient descent to find alignments between neural representations and symbolic variables, a technique further refined in [50] through Distributed Alignment Search (DAS), which allows individual neurons to participate in multiple causal roles. Such neuro-symbolic alignment is critical for diagnosing model failures; [18] highlights that while counterfactual interventions provide faithful evidence, they must be carefully designed to avoid biases arising from non-transitive dependencies and multiple sufficient causes.

Furthermore, the generation of causal explanations often requires synthesizing counterfactual data that adheres to structural constraints. [110] introduces a framework leveraging generative models and information-theoretic measures to ensure that perturbations in latent factors produce causally consistent changes in classifier outputs. This is complemented by [111], which utilizes causal generative learning to produce visually interpretable counterfactuals, offering superior explanatory fidelity compared to standard feature attribution methods. In the realm of sequence modeling, [112] employs variational autoencoders to generate meaningful input perturbations, constructing causal graphs over tokens to explain structured predictions.

Despite these advances, significant challenges remain in scaling these hybrid architectures and ensuring robustness against unobserved confounding. [20] argues for viewing neural architectures themselves as Structural Causal Models to compute feature effects from first principles, yet this requires strong assumptions about the underlying causal structure of input data. Moreover, [113] emphasizes the need for standardized evaluation metrics to compare different causal mediators and search methods systematically. Future research directions must focus on automating the discovery of causal abstractions without extensive human supervision and integrating large language models as proposers of causal factors, as suggested by emerging work in [90]. Ultimately, the synthesis of neural expressivity and symbolic rigor promises to transition deep learning from a correlational tool to a mechanistic reasoning engine capable of trustworthy deployment in high-stakes domains.

5 Causal Generative Modeling, Counterfactual Reasoning, and Effect Estimation

5.1 Deep Structural Causal Models for Counterfactual Synthesis

The integration of deep generative architectures with Structural Causal Models (SCMs) represents a pivotal advancement in achieving the third rung of Pearl's causal hierarchy: counterfactual reasoning. Unlike standard generative models that capture merely associational distributions, Deep SCMs are explicitly designed to perform the tripartite procedure of abduction, action, and prediction, thereby enabling the synthesis of high-dimensional data under hypothetical interventions while preserving causal consistency. A fundamental requirement for valid counterfactual inference is the ability to uniquely recover unit-specific exogenous noise variables from observations, a step known as abduction. To address this, recent methodologies have leveraged bijective transformations, particularly normalizing flows, which map complex observational data to simple latent noise distributions invertibly. [16] demonstrate that by employing normalizing flows within a variational framework, it is possible to achieve tractable inference of these exogenous variables, a capability often missing in non-invertible deep learning models. This invertibility is crucial; as shown in [15], autoregressive flow architectures naturally define a causal ordering over variables, allowing for the exact evaluation of counterfactual queries and ensuring that the learned mechanisms are identifiable under specific conditions.

Beyond flow-based approaches, the emergence of diffusion probabilistic models has opened newfrontiers for counterfactual synthesis in high-dimensional spaces such as images and videos. [114] propose Diff-SCM, a framework that utilizes iterative denoising processes guided by the gradients of anti-causal predictors to sample from counterfactual distributions. This approach effectively bridges the gap between the superior sample quality of diffusion models and the structural rigor of SCMs. Similarly, [115] introduce CausalDiffAE, which disentangles high-level semantic causal variables from stochastic variations, enabling granular control over interventions via do-operations in the latent space. These methods collectively address the challenge of generating realistic counterfactuals that respect the underlying causal graph, a task where traditional GANs often fail due to a lack of explicit mechanistic constraints.

The scalability of these models to complex, multi-variable systems has further been enhanced by modular architectures. [116] present a sequential training algorithm that decouples the learning of causal mechanisms from distribution estimation, allowing for the integration of pre-trained foundation models to answer identifiable causal queries even in the presence of latent confounders. This modularity is essential for applications in domains like healthcare, where [117] highlight the necessity of handling incomplete label records while maintaining causal fidelity. Furthermore, in dynamic settings, [118] illustrates the application of twin causal networks to generate counterfactual echocardiogram videos, demonstrating the practical utility of these models in simulating "what-if" scenarios for clinical decision-making.

Despite these advances, significant challenges remain regarding the identifiability of counterfactual distributions without strong assumptions. [119] establishes fundamental impossibility results, noting that counterfactual editing is theoretically impossible from i.i.d. samples alone without additional structural knowledge or relaxation strategies. Consequently, future research must focus on developing weaker identifiability conditions and robust estimation techniques that can operate under partial observability. The synthesis of bijective flows, diffusion processes, and modular designs suggests a converging trajectory toward causal foundation models capable of reliable, high-fidelity counterfactual synthesis across diverse modalities, ultimately transforming how deep learning systems reason about alternative realities.

5.2 Representation Learning for High-Dimensional Confounder Adjustment

While deep generative architectures and Structural Causal Models (SCMs) have unlocked the potential for counterfactual reasoning and high-fidelity synthesis, the foundational challenge of estimating causal effects from high-dimensional observational data remains rooted in mitigating confounding bias. Before advancing to the complex synthesis of alternative realities or the estimation of heterogeneous dynamic effects, it is imperative to address how latent or observed covariates influence both treatment assignment and outcomes. Traditional adjustment methods often falter in high-dimensional regimes due to the curse of dimensionality and the inadvertent inclusion of "bad controls"—such as colliders or mediators—that induce bias when conditioned upon. Deep representation learning has emerged as a potent paradigm to overcome these hurdles by mapping high-dimensional covariates into balanced, low-dimensional latent spaces that satisfy the unconfoundedness assumption while preserving predictive fidelity. A dominant strategy within this domain involves learning representations that render the treatment distribution invariant, effectively minimizing the discrepancy between treated and control groups. Shalit et al. [120] pioneered this approach by integrating domain adaptation metrics, such as Maximum Mean Discrepancy, into the loss function to enforce balance. Building on this foundation, [121] critiques the inherent trade-off between balance and prediction, proposing a synergistic framework that combines balancing weights with representation learning to target specific populations more accurately. Similarly, [122] extends classical matching by utilizing neural networks to derive multivariate balancing scores, overcoming the sparsity issues inherent in exact matching for high-dimensional inputs.

However, achieving balance on all observed covariates is insufficient and potentially detrimental if the covariate set includes instrumental variables or colliders. To mitigate this, recent works have shifted focus toward decomposing representations to isolate valid confounders from nuisance factors, a conceptual precursor to the disentangled strategies employed in continuous treatment settings. [123] proposes a synergistic framework that explicitly disentangles confounders from instrumental and adjustment variables, balancing only the former to prevent bias amplification. This decomposition is further refined in [124], which employs variational autoencoders to separate covariates into instrumental, confounding, and noise factors, laying the groundwork for more granular causal estimation. Addressing the specific challenge of "bad controls," [125] introduces Nearly Invariant Causal Estimation (NICE), which leverages Invariant Risk Minimization (IRM) to learn representations that strip out collider information while retaining sufficient statistics for valid adjustment. Complementing this, [126] utilizes mutual information maximization to preserve predictive information across treatment groups while filtering out irrelevant variables, thereby stabilizing estimates under severe covariate imbalance.

The complexity of confounding escalates when dealing with latent factors or temporal dynamics, bridging the gap between static representation balancing and the dynamic modeling required for longitudinal analysis. In scenarios where confounders are unobserved but proxied by high-dimensional data, [69] utilizes Variational Autoencoders to infer latent confounder summaries from noisy proxies, relaxing the strict no-unmeasured-confounding assumption. This is extended in [70], which incorporates unstructured data like images and text as proxies to recover latent confounders in multi-modal settings. For longitudinal data, [80] employs attention mechanisms to capture long-range dependencies among time-varying confounders, using a counterfactual domain confusion loss to learn balanced representations over time—a technique that foreshadows the transformer-based architectures used for dynamic treatment effect estimation. Furthermore, [127] addresses non-local confounding in spatial settings by learning representations of regional covariates to adjust for spatially dependent treatment assignments.

Despite these advances, significant challenges remain regarding the identifiability of latent confounders and the robustness of learned representations to model misspecification, concerns that echo the theoretical limitations discussed in the context of counterfactual synthesis. [76] warns that deep latent variable models may yield inconsistent causal estimates if the latent structure is misspecified, highlighting the gap between empirical success and theoretical guarantees. Moreover, [128] provides a refutation framework to quantify the bias introduced by dimensionality reduction, emphasizing that aggressive compression can violate the unconfoundedness assumption. Future research must therefore focus on developing methods that offer rigorous identifiability guarantees for latent confounders and integrate sensitivity analysis directly into the representation learning objective. By ensuring robustness against unmeasured confounding and model uncertainty, these foundational representation learning techniques will provide the stable basis necessary for the subsequent exploration of heterogeneous treatment effects and dynamic causal pathways.

5.3 Neural Estimation of Heterogeneous and Dynamic Treatment Effects

The estimation of heterogeneous and dynamic treatment effects represents a critical frontier where deep learning's expressive power addresses the non-linearities and high-dimensionality inherent in real-world causal inquiries. Traditional parametric methods often fail to capture complex dose-response relationships or time-varying confounding structures, necessitating architectures that can model intricate functional mappings while maintaining causal validity. A primary challenge lies in estimating the Conditional Average Treatment Effect (CATE) and continuous dose-response curves without introducing discontinuities or bias from selection mechanisms. Recent advancements have moved beyond simple stratification, employing varying coefficient neural networks to preserve the continuity of estimated average dose-response functions (ADRF) across continuous treatment spaces [129]. Furthermore, disentangled representation learning has emerged as a potent strategy, separating covariates into instrumental, confounding, and adjustment factors to isolate causal signals from noise, particularly in continuous treatment settings where balancing the entire representation space is suboptimal [124].

In scenarios involving sequential decision-making, such as personalized medicine or dynamic policy allocation, the complexity escalates due to time-dependent confounders that are themselves affected by prior treatments. Standard regression adjustments fail here, requiring methods that leverage the iterative G-computation formula within deep architectures. Models like DeepACE integrate recurrent structures with doubly robust estimation techniques to consistently estimate average causal effects from longitudinal patient trajectories, ensuring asymptotic efficiency even in the presence of complex temporal dependencies [130]. The advent of transformer-based architectures has further revolutionized this domain by capturing long-range dependencies among time-varying covariates and treatments. The Causal Transformer, for instance, utilizes cross-attention mechanisms to jointly model history-dependent states and employs counterfactual domain confusion losses to learn balanced representations that are predictive of outcomes yet invariant to treatment assignment [80]. Similarly, disentangled counterfactual recurrent networks (DCRN) explicitly decompose latent patient histories into distinct factors influencing treatment selection, outcomes, and confounding, thereby enhancing interpretability and forecast accuracy in dynamic regimes [131].

Despite these architectural innovations, the reliability of neural estimators hinges on rigorous uncertainty quantification and sensitivity analysis, especially when the ignorability assumption is suspect. Scalable sensitivity models have been developed to derive bounds on causal estimates under continuous interventions, accounting for potential hidden confounding that standard deep models might overlook [132]. Moreover, the integration of graph neural networks allows for the modeling of interference in multi-agent dynamical systems, where units are not independent but interact through underlying network structures. Approaches like CF-GODE extend causal inference to these interconnected settings by modeling continuous-time counterfactual outcomes using graph ordinary differential equations, effectively handling inter-unit confounding and dynamic treatment spillovers [133]. Looking forward, the field must address the tension between model flexibility and identifiability. While deep networks offer universal approximation, ensuring that learned representations satisfy sufficient conditions for causal identification remains an open theoretical challenge. Future research directions likely involve hybrid neuro-symbolic frameworks that embed structural causal constraints directly into the loss landscape, alongside advances in causal bandits for optimizing intervention strategies under general non-linear structural causal models [134]. Ultimately, the synergy of representation learning, temporal modeling, and robust inference protocols promises to deliver reliable, personalized causal insights from complex observational data.

5.4 Causal World Models for Planning and Explainable Reinforcement Learning

Building on the mechanistic world models required for robust reinforcement learning contexts, the integration of causal generative models into RL marks a paradigm shift from statistical correlation to mechanistic simulation, enabling agents to reason about interventions rather than mere associations. Unlike traditional model-based RL approaches that learn dense transition dynamics susceptible to spurious correlations, causal world models decompose the environment into invariant structural mechanisms. This decomposition is critical for robust planning, as it allows agents to generalize across distributional shifts by identifying and exploiting the underlying causal factors of variation. For instance, [135] demonstrates that Object-Oriented Causal Dynamics Models (OOCDM) can share causal parameters among object classes, significantly enhancing scalability and generalization in large-scale environments compared to non-causal baselines. Similarly, [136] introduces a structured world model that factorizes transition dynamics causally, allowing for modular adaptation to unseen environments by identifying sparse changes in the causal graph rather than relearning the entire state space.

A central challenge in high-dimensional planning is the computational intractability of modeling full observation spaces. Causal world models address this by constructing causally correct partial models that predict only the relevant subsets of observations necessary for decision-making. [137] theoretically establishes that standard partial models are often confounded by unmodeled variables, leading to erroneous planning; in contrast, their proposed framework ensures causal correctness by explicitly accounting for the confounding structure, thereby maintaining planning fidelity without full state reconstruction. This aligns with the findings in [138], which proves that learning state abstractions based on causal features yields policies that generalize to novel observations in block Markov Decision Processes (MDPs), whereas statistical abstractions fail under domain shifts. Furthermore, [139] provides a theoretical justification for this approach, showing that any agent satisfying a regret bound under a wide set of distributional shifts must implicitly learn an approximate causal model of the data-generating process.

Beyond robustness, causal world models offer a pathway to mechanistic explainability in RL. By tracing the influence of actions through learned causal graphs, agents can generate step-by-step justifications for their policies. [140] proposes training agents to build internal causal models of their own behavior, enabling them to generate semantically meaningful explanations through self-communication. This contrasts with post-hoc attribution methods that often rely on correlational heuristics. In the context of reward assignment, [141] utilizes generative causal models to decompose long-term returns into identifiable Markovian rewards, providing interpretable credit assignment that preserves policy invariance. Additionally, [142] introduces a framework for identifying functional actual causes in continuous RL settings, moving beyond heuristic proximity to principled causal attribution of specific events.

Despite these advances, significant challenges remain in scaling causal discovery to complex, partially observable domains and integrating these models with large-scale foundation models. Emerging trends suggest a convergence of causal discovery with active intervention strategies, where agents actively perturb the environment to refine their causal graphs, as seen in [143]. Future research must address the identifiability of latent causal variables in fully unsupervised settings and develop efficient algorithms for causal planning in continuous action spaces, ultimately bridging the gap between causal inference theory and practical, deployable RL systems.

6 Applications, Evaluation Frameworks, and Trustworthy Artificial Intelligence

6.1 Domain-Specific Deployments in Science and Industry

The deployment of causal deep learning in high-stakes domains marks a paradigm shift from predictive association to mechanistic understanding, addressing critical failures of correlational models under distribution shifts. In biomedical discovery and precision medicine, the integration of causal structural assumptions into deep architectures enables the disentanglement of confounding factors from true therapeutic effects. For instance, methods leveraging Variational Autoencoders to infer latent confounders from proxy variables have demonstrated superior robustness in estimating individual treatment effects compared to traditional propensity score matching [69]. Furthermore, the application of deep end-to-end frameworks capable of simultaneous causal discovery and effect estimation allows for the analysis of heterogeneous, mixed-type clinical data with missing values, bridging the gap between observational electronic health records and randomized control trial standards [144]. In medical imaging, causal analysis has clarified the suitability of semi-supervised learning strategies by explicitly modeling the causal generation of annotations, thereby mitigating risks associated with data mismatch and scarcity [145]. Recent advances in high-fidelity counterfactual synthesis using diffusion-based structural causal models further empower clinicians to simulate patient responses to unadministered treatments, facilitating "what-if" reasoning without invasive trials [114; 146].

In autonomous systems and robotic intelligence, causal representation learning is pivotal for achieving generalizable policies that distinguish between controllable actions and environmental nuisances. Agents equipped with causal curiosity mechanisms can autonomously design experiments to identify latent causal factors, significantly reducing the data required for learning complex dynamics compared to supervised planners [143]. By learning sparse, object-oriented causal dynamics, robots can perform task-independent state abstractions that generalize to unseen configurations, overcoming the brittleness of dense dynamics models prone to spurious correlations [147; 148]. Moreover, the fusion of observational and interventional data via latent-based causal transition models allows agents to safely leverage offline datasets generated by other agents, even in the presence of hidden confounders, enhancing sample efficiency in partially observable Markov decision processes [149].

Climate science and ecological modeling benefit from causal deep learning's capacity to handle high-dimensional, non-stationary time-series data where controlled interventions are impossible. Scalable sensitivity analysis frameworks now permit the estimation of continuous treatment effects, such as the impact of aerosol emissions on cloud properties, while quantifying uncertainty induced by unobserved confounders [132]. Similarly, causal graph neural networks have improved wildfire danger prediction by explicitly removing spurious links between highly correlated meteorological variables, ensuring models remain robust across different biome regimes [150]. In financial and economic contexts, causal machine learning addresses the "deconfounder" problem in multiple-cause settings, such as genome-wide association studies or market trend analysis, by inferring latent variables that substitute for unobserved confounders [75]. Additionally, the integration of multimodal data, including text and images, into double machine learning frameworks expands the scope of causal inference in economics and marketing, allowing for more nuanced effect estimation from unstructured data sources [151]. Despite these advances, challenges persist in validating causal assumptions in fully unsupervised settings and scaling these methods to massive, real-world graphs without compromising identifiability guarantees.

6.2 Quantitative Metrics for Causal Fidelity and Structural Validity

To bridge the gap between the diverse application successes outlined previously and the rigorous benchmarking ecosystems discussed subsequently, evaluating causal deep learning systems necessitates a paradigm shift from standard predictive accuracy to metrics that rigorously assess structural fidelity, representational disentanglement, and counterfactual validity. Unlike correlational models, causal algorithms must be judged on their ability to recover the underlying data-generating mechanisms, a requirement that demands specialized quantitative frameworks tailored to address the identifiability challenges inherent in real-world deployments.

The assessment of causal structure discovery primarily relies on graph-theoretic distances between the learned adjacency matrix and the ground truth. The Structural Hamming Distance (SHD) remains the dominant metric for quantifying skeleton and orientation errors; however, recent critiques highlight its sensitivity to scale and its inability to distinguish between Markov equivalent structures without interventional data [10; 11]. To address these limitations and better reflect the robustness required in dynamic environments, researchers increasingly employ the Mechanism Shift Score (MSS), which leverages heterogeneous environments to validate sparse mechanism changes, offering a more robust evaluation under the sparse mechanism shift hypothesis [152]. Furthermore, in low-data regimes common in high-stakes domains where point estimates are unreliable, Bayesian approaches utilize posterior probability mass over the true DAG or equivalence classes, as demonstrated by variational frameworks that outperform maximum-likelihood methods on SHD metrics [60].

Beyond graph topology, evaluating causal representation learning requires metrics that verify the alignment between latent variables and ground-truth causal factors, ensuring that the disentanglement achieved in domains like robotics and medicine is genuine rather than spurious. Traditional disentanglement scores often assume statistical independence, which fails to capture causally dependent latents; thus, new metrics based on “consistency” and “entanglement graphs” have been proposed to quantify partial disentanglement under mechanism sparsity regularization [37]. A critical advancement involves interventional consistency tests, where the model’s latent space is perturbed to ensure that changes propagate according to the learned causal graph rather than spurious correlations. For instance, the identifiability of latent factors under unknown interventions can be verified by analyzing the geometric signatures of the support set, providing a theoretical basis for empirical validation [8]. Moreover, the quality of representations is increasingly measured by their downstream transportability, assessing whether invariant features identified in source domains maintain predictive power under target distribution shifts, a capability central to causal domain generalization [153; 154].

The validity of generative causal models, particularly those used for counterfactual synthesis in clinical and economic settings, is uniquely assessed through counterfactual fidelity, measuring the semantic plausibility and minimality of generated samples under hypothetical interventions. Standard likelihood metrics are insufficient here; instead, axiomatic soundness measures evaluate whether the generated counterfactuals satisfy the three steps of abduction, action, and prediction inherent to Structural Causal Models [16; 146]. Recent work introduces specific distance metrics in latent space to ensure counterfactuals remain on the data manifold while adhering to causal constraints, alongside metrics that quantify the preservation of invariant mechanisms during generation [114]. However, a significant challenge remains in benchmarking, as high classification accuracy does not necessarily correlate with correct causal estimation, a phenomenon termed the “smoke and mirrors” effect in downstream tasks [155]. Consequently, the field is moving toward composite evaluation pipelines that integrate structural accuracy, interventional robustness, and counterfactual realism to serve as the foundation for the dataset and platform discussions in the next section. Future metrics must also account for the transitivity failures and multiple sufficient causes often missed by current counterfactual theories, ensuring that evaluative frameworks align with the nuanced realities of neural causal mechanisms [18].

6.3 Benchmark Datasets, Simulation Pipelines, and Evaluation Platforms

The empirical validation of causal deep learning methodologies necessitates a rigorous ecosystem of benchmark datasets, simulation pipelines, and evaluation platforms that bridge the gap between theoretical identifiability and practical applicability. While synthetic data generators provide the essential ground-truth causal structures required for algorithmic stress-testing, they must evolve beyond simple linear Gaussian models to capture the complexity of real-world systems. Advanced simulation frameworks now leverage normalizing flows and neural ordinary differential equations to produce high-dimensional data with controllable non-linear mechanisms and known confounding patterns, enabling the verification of identifiability claims under diverse conditions [156; 77]. For instance, specialized benchmarks like CausalTime offer time-series data where R^2-sortability artifacts are explicitly mitigated, ensuring that discovery algorithms are evaluated on genuine causal signals rather than statistical shortcuts [157]. Furthermore, semi-synthetic datasets constructed by injecting known causal effects into real-world distributions, such as electronic health records or image repositories, provide a critical intermediate testbed for assessing robustness against domain shifts while maintaining verifiable ground truth [158; 151].

Complementing synthetic resources, curated real-world datasets with partial or expert-annotated causal graphs are indispensable for evaluating the external validity of causal models. In domains like medical imaging and genomics, datasets such as those utilized in [145] and [16] allow researchers to test whether learned representations align with established biological mechanisms. The CauseEffectPairs benchmark remains a foundational resource for bivariate causal discovery, offering diverse cause-effect pairs from meteorology to economics to assess the ability of algorithms like Additive Noise Methods to distinguish directionality [159]. Moreover, emerging benchmarks specifically target higher-order causal reasoning, such as the ACRE dataset for abstract causal induction and the MuCR benchmark for multimodal causal reasoning in vision-language models, challenging systems to perform screening-off and backward-blocking reasoning beyond mere covariation [160; 161].

To standardize comparison and mitigate the fragmentation of evaluation protocols, unified benchmarking platforms and leaderboards are becoming increasingly vital. Frameworks like DoWhy-GCM extend existing libraries to support a wide range of causal questions, from structure learning to attribution, within a consistent graphical modeling interface [162]. Similarly, initiatives like CausalBench aim to provide reproducible experimental protocols that facilitate fair comparisons across diverse methodologies, addressing the critical challenge that predictive accuracy often fails to correlate with causal fidelity [155]. Despite these advancements, significant challenges persist in benchmark design, particularly regarding the risk of data leakage and the misalignment between proposed causal graphs and actual data generating processes. Future directions must prioritize the development of “shadow datasets” to prevent overfitting to specific benchmark characteristics and the creation of dynamic evaluation environments that test causal generalization under continuous distribution shifts, thereby ensuring that causal deep learning systems achieve true trustworthiness in high-stakes deployments.

6.4 Trustworthy AI through Causal Fairness, Robustness, and Interpretability

Building upon the rigorous evaluation ecosystems and benchmarking platforms established in the previous section, the integration of causal inference into deep learning architectures represents a paradigm shift from associative pattern matching to mechanistic reasoning, fundamentally enhancing the trustworthiness of artificial intelligence systems. By modeling the data-generating process through Structural Causal Models (SCMs), researchers can address the triad of fairness, robustness, and interpretability not as independent constraints, but as emergent properties of invariant causal mechanisms. In the domain of algorithmic fairness, causal frameworks transcend traditional associational metrics by distinguishing between legitimate correlations and discriminatory pathways. Approaches such as the Disentangled Causal Effect Variational Autoencoder (DCEVAE) enable the separation of sensitive attributes from latent representations, facilitating counterfactual fairness where predictions remain invariant under hypothetical interventions on protected variables [163]. Similarly, causally motivated shortcut removal techniques utilize auxiliary labels to enforce conditional independences implied by causal graphs, effectively mitigating bias arising from spurious correlations without sacrificing predictive utility [164]. This counterfactual perspective is further operationalized in natural language processing through frameworks like CausaLM, which fine-tune language models to estimate the true causal effect of high-level concepts, thereby isolating and removing unwanted biases ingrained in training corpora [109].

Robustness against distribution shifts and adversarial perturbations is similarly grounded in the Principle of Independent Causal Mechanisms. Standard deep models often fail out-of-distribution because they rely on environment-specific spurious features; conversely, causal representation learning seeks features that are invariant across domains. Theoretical analyses suggest that agents satisfying regret bounds under diverse distributional shifts must implicitly learn approximate causal models of the data-generating process [139]. Practical implementations, such as Causal Attention for Unbiased Visual Recognition, self-annotate confounders to learn attention masks that focus on causal features rather than background contexts, significantly improving performance in out-of-distribution settings [165]. Furthermore, generative interventions can be employed to manufacture synthetic data that breaks spurious correlations, steering representations toward causal consistency and enhancing generalization from source to target domains [166]. In the context of security, causal interventions via semantic smoothing have demonstrated provable robustness against natural language attacks by learning causal effects $p(y|do(x))$ rather than observational likelihoods [167].

Interpretability benefits profoundly from causal abstraction, which aligns low-level neural activations with high-level human-intelligible concepts. Unlike post-hoc attribution methods that may suffer from confounding, causal interpretability employs interchange interventions to verify whether neural representations faithfully realize a target causal model [48]. Techniques such as Distributed Alignment Search (DAS) overcome previous limitations by using gradient descent to find alignments between distributed neural representations and symbolic causal variables, revealing internal structures that brute-force methods miss [50]. Moreover, causal proxy models offer a pathway to faithful explanations by training surrogate networks that mimic black-box behavior while maintaining manipulable internal representations for counterfactual analysis [168]. Despite these advances, challenges remain in scaling causal discovery to high-dimensional data and defining appropriate mediators for complex non-linear systems [113]. Future research must focus on unifying causal representation learning with foundation models to create systems that are not only performant but inherently auditable, robust, and fair by design [169].

7 Conclusion

The convergence of causal inference and deep learning has transitioned from a theoretical aspiration to a methodological imperative, fundamentally reshaping how artificial systems perceive, reason, and act. This survey has delineated the trajectory from purely correlational models, which falter under distribution shifts and spurious dependencies, to causal representation learning frameworks that recover latent structural mechanisms. Key theoretical breakthroughs have established that identifiability of latent causal variables is achievable even in non-linear, non-parametric settings, provided there is sufficient diversity in observational environments or access to unknown interventions [7; 38]. These results dismantle the long-held belief that causal discovery is impossible without strong parametric assumptions, offering a rigorous foundation for unsupervised disentanglement [170].

Algorithmically, the field has witnessed a paradigm shift from discrete combinatorial search to continuous optimization and neural amortization. Differentiable structure learning methods now enable scalable graph discovery in high-dimensional spaces by relaxing acyclicity constraints into differentiable penalties [59], while amortized inference models learn to predict causal structures directly from data distributions, drastically reducing computational overhead [65; 14]. Furthermore, the integration of Large Language Models (LLMs) has emerged as a transformative frontier, where pre-trained linguistic knowledge serves as a structural prior to guide data-driven discovery, effectively bridging the gap between semantic understanding and statistical causality [84; 93]. However, critical analysis reveals that current LLMs often function as "causal parrots," reciting correlations embedded in training data rather than performing genuine causal reasoning, necessitating hybrid architectures that ground linguistic priors in empirical data [86].

Despite these advances, significant gaps remain. The distinction between expressivity and learnability persists; while neural networks can universally approximate structural causal models, learning the correct causal hierarchy from observational data alone remains theoretically impossible without specific inductive biases or interventional signals [3]. Moreover, existing evaluation frameworks often rely on synthetic benchmarks that fail to capture the complexity of real-world confounding and partial observability, hindering the deployment of causal models in high-stakes domains [171]. Future research must prioritize the development of causal foundation models that internalize the principles of independent causal mechanisms, enabling zero-shot adaptation to novel environments and robust counterfactual reasoning [5]. This requires a synergistic approach combining neuro-symbolic integration, where the perceptual power of deep learning is constrained by the logical rigor of symbolic causal calculus [53; 48]. Ultimately, realizing the full potential of causal deep learning demands not only algorithmic innovation but also the establishment of standardized benchmarks and ethical guidelines to ensure that these systems provide trustworthy, interpretable, and fair decisions in complex societal applications [172].

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